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How TikTok leads users to flow experience: investigating the effects of technology affordances with user experience level and video length as moderators

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Abstract

Purpose – Although the short-video-based application TikTok and its AI-enhanced technology have achieved enormous success and reshaped the user experience, few studies have focused on the user experience in the TikTok context. This study adopts a technology affordance theory lens to identify the main mechanisms contributing to the user experience in the short-video platform context while including user experience level and video length as moderating effects.

Design/methodology/approach – This study collected 401 valid questionnaires from TikTok users and used the structural equation modeling approach to examine the proposed research model.

Findings – Three technology affordances (perceived effortlessness, perceived recommendation accuracy, and perceived recommendation serendipity) contribute to TikTok users' optimal flow experience. Multi-group analysis results further indicate that individuals react differently to the same stimuli as their experience level increases. Finally, video length critically influences the impact of technology affordances on users' cognitive responses.

Originality/value – As a burgeoning industry, the mechanisms enabling short-video platforms to engage users remain unclear to practitioners and researchers. Thus, this study's technology affordance lens provides necessary insights into how TikTok's innovative and advanced technologies contribute to user flow experience from a context-dependent perspective. Furthermore, given that most existing studies have neglected possible variations in user preferences when investigating the effects of technology, this study enriches the existing literature by employing user experience level and video length as moderators.

Keywords TikTok, Technology affordance, User experience level, Flow experience, Moderator

Paper type Research paper

1. Introduction

TikTok is a video-sharing mobile application that encourages users to upload short (usually under 3 min) videos with built-in special effects, filters, and stickers. Since its introduction, TikTok has enjoyed dramatic growth, quickly becoming one of the world's most popular social media (Geyser, 2021) and cultivating receptive and engaged communities around its videos. TikTok features unique content-delivery and -discovery mechanisms that distinguish it from traditional platforms. For instance, TikTok has no homepage or start button, its videos play automatically when the app is opened, and users can only view content as curated by the recommendation algorithms. These innovations have led to TikTok's unprecedented

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success, especially in terms of engaging and retaining users. Some studies have revealed that TikTok surpasses the social media giants (i.e. Facebook and Instagram) and longer-video platforms (e.g. YouTube) in time spent per user (Price, 2020), a widely-used metric for user engagement level (Wu *et al.*, 2018; Atkins *et al.*, 2017). In addition, App Annie (2021) reported that time spent on TikTok increased by 325% in 2020, with the monthly time spent by each TikTok user also growing faster than that of users of any other app. Because engaging users is a priority for designers of products and services of all kinds (Doherty and Doherty, 2018), understanding the mechanisms that contribute to TikTok's success can provide useful guidelines for designers and practitioners in the short-video industry (Zhang *et al.*, 2019) and influence traditional social media to develop new strategies to engage and retain users.

Engagement has been conceptualized by Csikszentmihalyi (1990) using flow theory (Elias *et al.*, 2010), which describes flow as a state in which users are fully engaged and immersed in a given activity. Flow experience has been widely adopted in information systems (IS) research to reflect the depth of user engagement (Suh *et al.*, 2017). Therefore, flow represents an appropriate lens for investigating and operationalizing the mechanisms responsible for the profound user engagement with TikTok (Suh *et al.*, 2017; Rupayana, 2008). However, the existing flow experience literature insufficiently explains the TikTok user experience due to disagreement and inconsistencies surrounding the conceptualization and antecedents to flow in different contexts (Valinatajbahnamiri and Siahtiri, 2021; Triberti *et al.*, 2021). According to Hoffman and Novak (2009), once a new technology has been introduced, the antecedents and applications of flow should be re-examined using a context-specific approach. Given the unique interaction behavior associated with TikTok, the current study investigates the mechanisms that explicitly contribute to user experience in the TikTok context to extend flow theory in this domain.

Furthermore, various studies have highlighted the difficulty of sustaining user engagement because of the short-lived impact of some elements (Suh *et al.*, 2017). This problem produces a critical challenge for online communities entering their mature stage, namely, maintaining user interest as user experience with the online community increases (Fu, 2019). Notably, few studies have considered the possibility that the same activity will not induce a consistent flow experience in a given user as their experience increases. Accordingly, the current study introduces user experience level—describing the time since a user first used a given technology (Shao *et al.*, 2020)—as a moderator in the relationship between technological affordances and flow experience. As such, our study not only extends flow theory beyond the limited person-related perspective but also identifies the factors responsible for retaining users long-term. Our findings may alert other platforms to the fact that the factors that initially attract users may be distinct from those that retain them long-term, demanding the engagement of dynamic strategies as the online community grows.

This paper is organized as follows. First, we explain the theories of flow experience and technology affordances and discuss the development of the research model. Second, we detail the research methodology and present the data analysis results. Finally, we discuss the paper's findings and contributions.

2. Literature review and research hypotheses

2.1 Flow experience

Flow experience refers to "the holistic sensation that people feel when they act with total involvement." The construct was introduced by the psychologist Csikszentmihalyi (1975) following interviews with rock climbers, chess players, and dancers. According to Csikszentmihalyi (1975, 1988, 1997), in a flow state, people become fully immersed in a given activity, and the experience itself is so enjoyable that changes in their surroundings go unrecognized. Csikszentmihalyi (1990) identifies nine conditions for the occurrence of flow: challenge-skill balance, clear goals, immediate feedback, concentration, loss of self-consciousness,

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sense of control, time distortion, an experience of becoming autotelic, and the merging of action and awareness. However, given that it was not clear whether all of these conditions applied to contexts beyond the original flow activities considered by Csikszentmihalyi, subsequent research has conceptualized and operationalized flow in other contexts. For example, flow experience has received considerable interest from researchers investigating computer-mediated environments, with Hoffman and Novak (1996) proposing that user flow experience provides a new research framework for investigating and understanding user behavior in such technologically augmented environments. The extent to which a new interactive technology can lead users to flow experiences can indicate the quality of that technology (Triberti *et al.*, 2021).

In the IS field, researchers have investigated flow in terms of its antecedents (or drivers) and dimensions (Trevino and Webster, 1992; Webster *et al.*, 1993; Hoffman and Novak, 1996; Koufaris, 2002; Zhang *et al.*, 2014; Suh *et al.*, 2017; Lu and Cheng, 2020). Table 1 summarizes 19 representative articles that investigate flow from the perspectives of antecedents, moderators, and dimensions of the flow experience. Overall, we observed that although the antecedents to flow vary in different research contexts, the lenses adopted are normally artifact-related (e.g. Suh *et al.*, 2017; Martins *et al.*, 2019; Gao *et al.*, 2017; Khan *et al.*, 2017; Carlson and O'Cass, 2011), person-related (e.g. Jeon *et al.*, 2018; Rose *et al.*, 2012; Carlson and O'Cass, 2011; Khan *et al.*, 2017; Ghasemaghaei, 2019; Sanchez-Franco, 2006), and context-related (Trevino and Webster, 1992; Ghani *et al.*, 1991; Sreejesh *et al.*, 2021). Therefore, to examine flow experience in the TikTok context, we have also adopted these three factors as an overarching framework for investigating flow in such technologically augmented environments. Table 1 also facilitates the observation of gaps in the literature.

First, the literature reveals considerable inconsistency regarding flow dimensions, something previous researchers have recognized. For instance, while some studies have adopted all of Csikszentmihalyi's nine dimensions (e.g. Siemens *et al.*, 2015), others have employed fewer dimensions—whether two (e.g. Chen *et al.*, 2018), four (e.g. Jeon *et al.*, 2018), or five (e.g. Drengner *et al.*, 2018)—depending on the nature of the research and the overlap of concepts (e.g. concentration and immersion) between the different dimensions (Valinatajbahnamiri and Siahtiri, 2021). New technologies have led researchers to challenge the traditional concept of flow (Triberti *et al.*, 2021). For example, the internet-based learning context has been considered in terms of three dimensions (temporal dissociation, focused immersion, and heightened enjoyment), with curiosity and control excluded because users have neither total control over the task nor a variety of activities to explore (Saadé and Bahli, 2005). Therefore, the existing literature encourages future research to consider contextual factors to identify the most relevant conditions for computer-mediated environments (Park, 2014), leading relevant studies to subsequently structure flow using a context-specific approach, with Lu and Cheng (2020) integrating time distortion, concentration, curiosity, and enjoyment in the context of an online video platform (namely YouTube) environment. Because that study is the most relevant to our research context, the current study incorporates these four dimensions to examine flow.

Second, as Table 1 shows, of the antecedents to flow, artifact-related factors play a more prominent role because "the characteristics of the artifact itself may influence the likelihood of an optimal experience" (Finneran and Zhang, 2003). According to the literature, artifact-related factors dominate any examination of the antecedents to flow for a new technology. Artifact refers to information technology, computers, or any type of software application used by individuals to perform a task (Finneran and Zhang, 2003). Substantial research has adopted technology affordance theory to identify artifact-related factors when investigating user flow experience (Park, 2014; Shao *et al.*, 2020). From the table, we can observe that although the antecedents to flow experience have been extensively studied, most studies have focused on artifact-related factors from a context-specific perspective (e.g. competition affordance in the gaming context; sociability in the context of social network sites; and

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Study	Contexts	Flow dimension	Antecedents	Moderators
Ghani *et al.* (1991)	Computer-mediated environments	Enjoyment and concentration	Individual skills (P) Perceived challenge (O) Face-to-face groups vs. computer-mediated groups (C)	No moderator
Trevino and Webster (1992)	Computer-mediated environments	Control, attention focus, curiosity, and intrinsic interest	Ease of use (A) Computer skill (P) Electronic mail vs. voice mail (C) Communication partners' medium use (C)	No moderator
Webster *et al.* (1993)	Human-computer interactions	Control, attention focus, curiosity, and intrinsic interest	Flexibility/ Modifiability (A) Experimentation(A) Voluntary use (O)	No moderator
Hoffman and Novak (1996)	Computer-mediated environments	Involvement, attention focus, and telepresence	Skill (P) Challenge (O) Interactivity (A) Vividness (A) Involvement (A) Telepresence (A)	No moderator
Koufaris (2002)	Online shopping	Perceived control, enjoyment, and concentration	Product involvement (O) Web skills (P) Value-added search mechanisms (A) Challenges (O)	No moderator
Saadé and Bahli (2005)	E-learning	Temporal dissociation, focused immersion, and heightened enjoyment	Perceived usefulness (P&A) Perceived ease of use (P&A)	No moderator
Sanchez-Franco (2006)	Websites	Enjoyment and concentration	Perceived usefulness (P&A) Perceived ease of use (P&A)	Age (P)
Carlson and O'Cass (2011)	Websites	Time distortion, control, and involvement	Website technical performance (A) Website aesthetic performance (A) Website exchange performance (A) Website communication performance (A)	Internet self-efficacy (P)
Rose *et al.* (2012)	Online shopping	Telepresence, challenge, control, and attention focus	Interactive speed (A) Telepresence (A) Challenge (O) Skill (P)	No moderator

(*continued*)

Table 1. Literature review: flow experience in the information systems field

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Study	Contexts	Flow dimension	Antecedents	Moderators
Zhang *et al.* (2014)	Social network sites	Perceived sense of control, attention focus, curiosity, and intrinsic interests	Perceived interactivity (A) Perceived personalization (A) Perceived sociability (A) Social presence ability of SNS (A)	No moderator
Siemens *et al.* (2015)	Computer games	Challenge-skill balance, clear goals, immediate feedback, concentration, loss of self-consciousness, control, time distortion, autotelic experience, and the merging of action and awareness	Progress visibility (A)	Feedback type (O)
Gao *et al.* (2017)	Virtual worlds	Enjoyment, concentration, and perceived control	Information quality (A) System quality (A)	No moderator
Khan *et al.* (2017)	E-learning	Enjoyment and absorption	Perceived expertise (P) Perceived similarity (P&A) Perceived familiarity (P&A) Informativeness (A) Effectiveness (A)	No moderator
Suh *et al.* (2017)	Gamification	Immersion, time distortion, control, and enjoyment	Rewards affordance (A) Status affordance (A) Competition affordance (A) Self-expression affordance (A)	No moderator
Jeon *et al.* (2018)	Websites	Enjoyment, concentration, time distortion, and control	Informativeness (A) Design (A) Interactivity (A)	Users' personality (P)
Ghasemaghaei (2019)	Online shopping	Cognitive absorption	In-depth recommendation agent (A)	Product knowledge (P) Gender (P)
Martins *et al.* (2019)	Mobile service	Control and focused attention	Informativeness (A) Credibility(A) Entertainment (A) Irritation (A) Incentives (A) Web design quality (A)	No moderator
Lu and Cheng (2020)	Online video platforms	Time distortion, concentration, curiosity, and enjoyment	Induced curiosity (A) Perceived serendipity (A)	No moderator
Sreejesh *et al.* (2021)	In-game advertising	Cognitive absorption, time distortion, and enjoyment	PC-based (vs. mobile-based) in-game advertising (C) Accessed from social media (vs brand web) (C)	No moderator

Table 1.

Note(s): (A) refers to artifact-related factors, (P) refers to person-related factors, (C) refers to context-related factors, and (O) refers to other factors

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informativeness in the e-learning context). As such, the existing flow experience literature insufficiently explains the TikTok user experience due to a combination of the inconsistencies surrounding antecedents to flow in different contexts and the relative novelty of the application (Valinatajbahnamiri and Siahtiri, 2021; Triberti *et al.*, 2021). Thus, the characteristics of TikTok that make the platform stand out from other platforms—in terms of engaging users—remain unknown. Without this knowledge, it is difficult to understand how TikTok effectively induces the flow experience and thus to inspire other short-video platforms to develop fruitful strategies to engage users.

Third, although different artifact-related factors have been discussed in the literature, how individuals evaluate them differently—especially as their experience increases—has not been explained. Most studies have ignored how person-related factors impact the flow experience and moderate the effects of artifact-related factors. Moreover, the relevant existing studies have focused on static person-related factors (e.g. personal traits, gender, and age) that are enduring and relatively invariant to situational stimuli (Li and Browne, 2006). Notably, this focus complicates the ability of platforms to detect the reasons for losing users. Furthermore, most previous studies have neglected the possibility that the same activity may not consistently induce the flow experience for the same person as their experience with a specific technology increases. According to Valinatajbahnamiri and Siahtiri (2021), person-related factors that reflect dynamic individual states remain largely under-investigated. Therefore, the current study introduces the user experience level as a moderator in the relationship between technological affordances and flow experience.

Last, in addition to artifact-related and person-related factors, the existing literature encourages future work to consider contextual factors to identify the most effective flow-inducing factors for computer-mediated environments (Valinatajbahnamiri and Siahtiri, 2021). To reflect TikTok's unique short-video format, the current study introduces the context-related factor of video length. The following sections discuss the three factors in detail.

2.2 Antecedents to flow

2.2.1 The effects of technology affordances. The concept of affordance constitutes an appropriate theoretical lens for investigating the influence of technology on a user's psychological response (Shao *et al.*, 2020). Affordances were first proposed by Gibson (1986) to refer to the action possibilities enabled by an environment in an ecological psychology context. Given affordance theory's substantial influence on the understanding of behavioral intentions, it has been introduced to various disciplines, including psychology, media and communications, design, and IS. Hutchby (2001) first applied affordance theory in an IS context by transposing environmental concepts onto technological constructs. Thus, "technology affordance" describes a possibility for action enabled by a technological artifact, providing an appropriate lens through which to examine the influence of artifact-related factors on the TikTok user experience. Elsewhere, Suh *et al.* (2017), Strong *et al.* (2014), and Park (2014) have also argued that affordance theory offers an analytical link between technology features and user flow experience. Therefore, following the existing literature, the current study adopts the affordance theory lens to understand artifact-related factors of TikTok.

Given that technology differs in different contexts, no standard classification method has been identified from the literature. Several studies have recognized that affordances must be triggered (Volkoff and Strong, 2013) or identified (Strong *et al.*, 2014) by a goal-oriented actor to achieve an outcome enabled by a technology. Meanwhile, for Gaver (1991), affordances should be investigated by understanding both the capabilities and limitations of technology (i.e. knowing the probabilities they offer) and the needs (or goals) of prospective users. Elsewhere, Greeno (1994) indicates that affordance perception describes the process of an

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actor recognizing a technology's features and their own goals. Given that the main goal of TikTok users is to view interesting content (Statista, 2020; Frăţilă, 2021), we have understood affordance in terms of content-related artifacts, namely, what content is presented to users, and how it is presented. To capture the first element, we focus on TikTok's smart recommendation system, recognizing that TikTok content is fed automatically to users through its recommendation algorithms. This choice enables this study to build on the recommendations in the literature on algorithm-related affordances, including perceived recommendation accuracy, and considerations drawn from discussions of the overfitting problem, including introducing recommendation serendipity (Matt *et al.*, 2014). To capture how content is presented to users, we have identified perceived effortlessness, which refers to the extent to which users believe that they can interact with a product or application with little or no cognitive load (Diefenbach and Ullrich, 2015), reflecting the product's simplicity and intuitiveness. Perceived effortlessness, which reflects the effort expectancy of using a technology (Agarwal *et al.*, 2009), has also been captured as perceived ease of use (Wu *et al.*, 2008), which describes "the degree to which a person believes that using a particular system would be free from effort" (Davis, 1989). The two constructs are sometimes used interchangeably to reflect how free of effort using a technology is (Bankole *et al.*, 2012).

Specifically, in terms of what content is presented to users, a much-discussed topic related to TikTok's technology is its AI-based recommendation algorithm, considered by many the secret to the app's success (Financial Times, 2020; Alexander, 2020a). TikTok's recommendation system more significantly determines user experience than search-oriented mechanisms because the application automatically serves users recommended content. TikTok aggressively dictates what a user sees, precluding individual choice and predominantly determining the TikTok user experience. Following the literature, we have identified two recommendation-algorithm-related affordances: perceived recommendation accuracy and perceived recommendation serendipity.

Perceived recommendation accuracy, which describes the degree to which recommended content is accurately customized to a user's preferences, represents the focus of most relevant studies (Wang *et al.*, 2020; Roudposhti *et al.*, 2018; Zhu *et al.*, 2018). TikTok's famously hyper-accurate algorithms have attracted substantial attention. However, few details are public, with the company maintaining the technology as proprietary and confidential. In 2020, TikTok officially disclosed that its recommendation engines rely on very complex factors to provide each user with a personalized and concise feed. In addition to historical usage data, certain peripheral data—such as user device and account settings—are also considered indicators of interest that can precisely identify user preferences (TikTok Newsroom, 2020). The existing literature (Ma *et al.*, 2021; Kim and Han, 2014) suggests that limited user attention means that users preferentially allocate attention to the content that matches their preferences. According to flow theory, one dimension of flow experience in the social media context is concentration (Webster *et al.*, 1993; Lu and Cheng, 2020), a process that is likely to produce positive flow experiences and that can presumably be applied to the short-video platform context. Elsewhere, the content preferred by users is more likely to lead users to an enjoyable and interesting experience, another dimension of flow. Hence, we assume that the recommended content and its accordance with user preferences promotes a positive flow experience, assuming the content provokes focused attention and enjoyment (Ma *et al.*, 2021), leading to this study's first hypothesis:

H1. On short-video platforms, perceived recommendation accuracy is positively associated with the flow experience.

Nonetheless, when an algorithm becomes too accurate, an "information cocoon" trap may occur. This describes a phenomenon in which people, when faced with large amounts of online information, tend to see only what they want to see, with the algorithm selecting their

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preferred information, like a silkworm making a cocoon for itself (Zuiderveen Borgesius *et al.*, 2016). This process can occur in the TikTok context, with TikTok's official website (TikTok Newsroom, 2020) describing it as the "filter bubble" or "overfitting," which occurs when the recommendation engine is increasingly personalized, leading users to be presented with a narrow range of content. Subsequently, the accuracy-focused recommendation engine risks re-directing user experience, with existing studies offering a critique of the design focus on improving recommendation accuracy regularly at the cost of serendipity and diversity (Jin *et al.*, 2018). That is, a good recommendation system should be able to not only identify the content preferred by users but also to continually present unexpected content to prevent overfitting (Kunaver and Požrl, 2017). Following the introduction of recommendation serendipity as a possible solution to this problem (Matt *et al.*, 2014), we have introduced perceived recommendation serendipity to our study to reflect the extent to which users believe that the recommended videos exceed their expectations and trigger feelings of surprise (Liang, 2018; Lu and Cheng, 2020). Such serendipitous experiences are desirable because they evoke the "surprise and delight" feelings that heavily influence flow experience (Lu and Cheng, 2020). Given that a user's sense of curiosity and enjoyment have been identified as significant dimensions of flow, especially in the social media context (Webster *et al.*, 1993; Lu and Cheng, 2020), unexpected recommendations have been proposed as influencing user flow experience. According to Chen *et al.* (2018) and Liu (2017), random and unexpected factors are favorable in gamified settings because a certain degree of uncertainty drives users to become deeply immersed in an activity, a phenomenon that TikTok takes advantage of. Unlike traditional video platforms, in which users represent active information seekers, short-video platforms automatically feed content to their passive information receivers. Accordingly, users are served endless random videos, never knowing what the next video is going to deliver. This unpredictability likely evokes surprise and delight, encouraging an optimal flow experience. Hence, given that serendipity contributes to user flow experience, following the existing literature (Lu and Cheng, 2020), we propose that the degree to which users believe that the videos recommended to them are serendipitous positively influences flow, leading to our second hypothesis:

H2. On short-video platforms, perceived recommendation serendipity is positively associated with the flow experience.

Meanwhile, considering how its content is consumed by users, TikTok's extremely simple and intuitive operation process represents another critical component of the platform's success. Unlike most similar platforms, TikTok automatically begins playing the recommended short videos while a user browses, not impacting a user's cognitive load by making them decide what to watch based on a lengthy suggestion list. Additionally, users can easily scan through different videos by simply scrolling up and down (Wang, 2019). According to Krug (2005), the first law of designing a good technology artifact is "don't make me think," meaning users should be able to accomplish their intended tasks as intuitively and effortlessly as possible. Thus, lightning the cognitive load enhances the user experience. As such, we propose that TikTok's success depends substantially on the effortlessness of its operational process, leading this study to include the technology affordance of perceived effortlessness to refer to the extent to which a user believes that they can interact with a product or application without sacrificing substantial cognitive load (Diefenbach and Ullrich, 2015). We propose that user flow experience is largely determined by perceived effortlessness because a simple operation process enables users to concentrate on the content itself (Webster *et al.*, 1993; Lu and Cheng, 2020). In contrast, when increased cognitive effort is required, users are easily distracted from an activity, and their cognitive capacity is more likely to be overloaded, ultimately inducing frustration and anxiety and removing users from the flow experience. As such, we propose that TikTok users perceive an effortless operating process

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that significantly impacts flow experience. Researchers have also noted the critical influence of perceived effortlessness on user flow experience (Harris *et al.*, 2017; Zhou, 2012), a notion that has led to the third hypothesis:

H3. On short-video platforms, perceived effortlessness is positively associated with the flow experience.

2.2.2 The moderating effects of user experience level. As mentioned, person-related factors cannot be overlooked when investigating the factors that influence user flow experience. These factors are based on the differences among individuals that may yield different flow levels from the same task over time (Smith and Sivakumar, 2004). Although various studies have focused on personal traits, the other dimension of person-related factors, the dynamic state (Woszczynski *et al.*, 2002), remains largely under-investigated. Dynamic state attributes vary according to the particular situation and can include the mitigation by user experience level. To capture this attribute, the current study considers user experience level in the TikTok context.

As mentioned, user experience level describes the time since a user first used a given technology (Shao *et al.*, 2020). As noted by the psychologist Helson (1964), individuals react differently to the same stimuli as their experience increases; this is known as the adaptation level theory. Elsewhere, Sundar (2008) posits that certain technology affordances dazzle users at first but fail to sustain long-term interest, with individuals reacting differently to the same stimuli as their experience level increases (Helson, 1964). Following the literature, we propose that the extent of user experience also impacts the strength of the effect of technology affordances on user cognitive responses in the short-video platform context.

Specifically, we propose that, as experience level increases, the effect of perceived recommendation accuracy weakens and the effect of perceived recommendation serendipity strengthens. The reason is that, according to adaptation level theory, people gradually adapt to a new situation, leading to that situation becoming the norm (Helson, 1964), and demand another new experience. Elsewhere, according to Vafeas and Hughes (2021), "In our all-too-human lives, nothing maintains infinite momentum. We lose interest. And our love can fade as time goes by. The relationship fails to remain interesting—the belief is that there is nothing new. It might start to look like a lot of sameness." This argument also applies to the relationship between TikTok and its users. Although hyper-specific recommendations are accurate and constantly recommend content that interests users (e.g. sports, pets, and beauty), people become bored easily, meaning that super-accurate recommendations might not work for long-term users. As a user's experience on a platform increases, the preferred content of the user is likely to be narrowed down by the hyper-accurate recommendation algorithm. Thereafter, repetition—which implies a lack of variety—can lead users to experience boredom, ultimately reducing their interest in and engagement with the platform (Cummings *et al.*, 2016). Accordingly, we argue that the effects of perceived recommendation accuracy decrease as user experience level increases. Therefore, we propose that user experience level weakens the influence of perceived recommendation accuracy on flow experience. Recommendation serendipity has been proven a possible solution to this problem (Matt *et al.*, 2014), with adaptation level theory claiming that when users become bored with a current situation, they increasingly require new and unexpected experiences. Given that serendipity reflects unexpected and surprising experiences, serendipity's influence should increase as user experience level increases, leading to the following hypotheses:

H4. On short-video platforms, user experience level weakens the relationship between perceived recommendation accuracy and flow experience.

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H5. On short-video platforms, user experience level strengthens the relationship between perceived recommendation serendipity and flow experience.

Elsewhere, we propose that the effect of perceived effortlessness decreases as user experience with an online community increases. The literature provides evidence that perceived ease of navigation in the blogging context is more critical at the initial usage stage (Lu and Lee, 2012). This finding also applies to TikTok, because the effort required to complete a task matters more for low-experience users, who are unfamiliar with the operation and navigation process of a platform, than for high-experience users, who are familiar with the process. Again, according to adaptation level theory, although the simple operation process dazzles users at first, they become accustomed to it as their experience increases, taking it for granted and demanding something new. After that, it makes sense to argue that perceived effortlessness more substantially determines the cognitive response for low-experience users than for high-experience users. This notion led to the development of the sixth hypothesis:

H6. On short-video platforms, user experience level weakens the relationship between perceived effortlessness and flow experience.

2.2.3 The moderating effect of video length. In terms of context-related factors, we propose that video length critically influences the impact of technology affordances on user cognitive responses. TikTok's popularity is substantially due to its short-format content, which likely relates to people's increasingly short attention spans (Lin *et al.*, 2020). Thus, short-format content might contribute to the technology's effects, especially in terms of how the content is presented. We assume that video length weakens the impact of perceived effortlessness on flow experience because shorter content requires higher-frequency browsing, which demands a greater perceived effortlessness effect compared to browsing longer-form content. Therefore, significantly greater perceived operating simplicity should be demanded for shorter-format platforms. For example, compared to 3-min videos, browsing different 15-s videos requires 12 times more operational processing, which can make it easier for users to escape the flow (due to the complexity of the operating process). Thus, perceived effortlessness will play a much stronger role in determining user flow experience when video length is shorter. This notion led to the development of the following hypothesis:

H7. On short-video platforms, video length weakens the relationship between perceived effortlessness and flow experience.

Based on the discussion above, we propose the research model shown in Figure 1.

3. Research methodology

We adopted a cross-sectional survey to test the proposed research model and used a partial least squares (PLS) approach (using SmartPLS 3.0) for data analysis. The following sections detail the development of the measurement items and the data collection process.

3.1 Measurement items

The operationalized instruments were mainly adapted from the existing literature and modified to fit the TikTok context. As Table 2 shows, items for perceived recommendation accuracy were adapted from Roudposhti *et al.* (2018) and Knijnenburg *et al.* (2012). Perceived recommendation serendipity was measured based on items developed by Lu and Cheng (2020), and items for perceived effortlessness were adapted from Yu and Kong (2016). To assess flow experience, we borrowed items from Lu and Cheng (2020) because the context of those items was most relevant to that of our study. All items were measured using a

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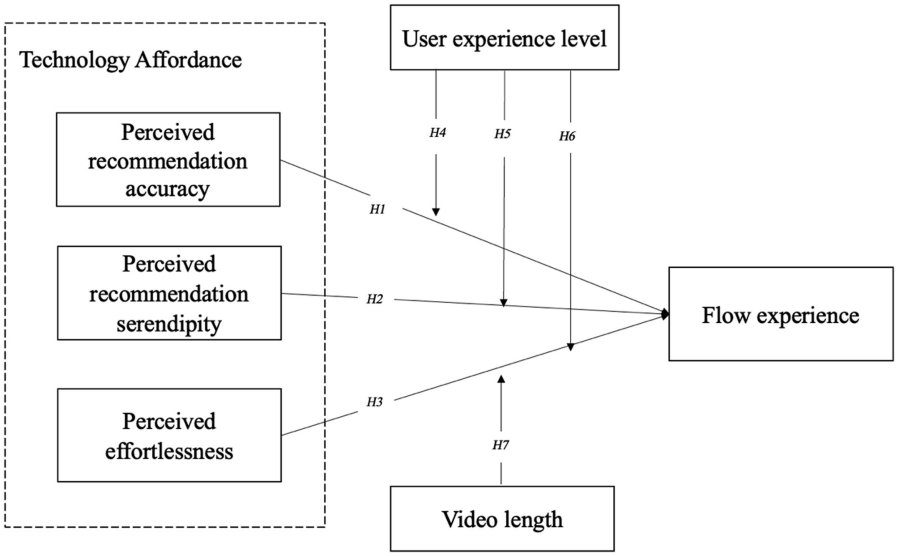


Figure 1.
Research model

Constructs	Items	References
Perceived recommendation accuracy (PRA)	PRA1-PRA3	Roudposhti *et al.* (2018), Knijnenburg *et al.* (2012)
Perceived recommendation serendipity (PRS)	PRS1-PRS4	Lu and Cheng (2020)
Perceived effortlessness (PE)	PE1-PE2	Yu and Kong (2016)
Flow experience (FE)	FE1-FE4	Lu and Cheng (2020)

Table 2.
Constructs and references

seven-point Likert scale allowing responses ranging from "strongly disagree" (1) to "strongly agree" (7).

Additionally, following the previous literature, we measured user experience on a single ordinal scale that assessed the passage of time from a respondent's initial use of TikTok (Shao *et al.*, 2020) and measured video length using an ordinal scale that matched its evolving stages. Notably, although TikTok expanded the video length constraint to three minutes for all users in July 2021, TikTok gave some users (especially influencers) early access to the possibility of uploading videos as long as three minutes in December 2020, henceforth allowing users to consume videos longer than one minute (Alexander, 2020b). According to TikTok product manager Drew Kirchhoff, "Some of you might have come across a longer video on TikTok already—we've been letting creators around the world experiment with the expanded format (i.e. up to 3 min) to give them more time for cooking demos, beauty tutorials, educational lesson plans, and comedy sketches" (Flannery, 2021).

A list of specific items and corresponding references is presented in Table A1. Additionally, following Shao *et al.* (2020), age, gender, and the highest education level are used as control variables.

3.2 Data collection

Data were collected from TikTok users by distributing questionnaires on Amazon Mechanical Turk (MTurk) between February 19 and February 25, 2021. A

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pre-qualification question, "Are you a user of TikTok?", was used to exclude respondents with no experience with the application, ensuring that only data from TikTok users would be processed.

The quality of the data provided by MTurk samples is quite high, typically equaling the quality obtained using traditional college student samples (Buhrmester *et al.*, 2011; Farrell *et al.*, 2017; Goodman *et al.*, 2013; Horton *et al.*, 2011). Some studies have used MTurk to effectively identify social media (e.g. Facebook and TikTok) users via self-reported questions (e.g. are you a Facebook/TikTok user?) (Zheng *et al.*, 2020; Tang, 2021). Given the verified quality of self-reported data collected using MTurk, we have also adopted MTurk as an acceptable data collection approach.

A total of 401 valid samples were identified after deleting incomplete data, data from non-TikTok users, and data from users who responded that they watch videos longer than 3 min, which would indicate either invalid samples or users using the Chinese version of TikTok. Demographic information is summarized in Table 3.

4. Results

4.1 Testing the measurement model

To test the measurement model, we examined both the internal reliability and convergent and discriminant validity of each construct's items. As Table 4 shows, factor loadings for each construct were all above 0.7, suggesting acceptable convergent validity (Barclay *et al.*, 1995; Hulland, 1999). Internal reliability was assessed by examining Cronbach's alpha, rho_A, and the composite reliability values of each construct. All relevant values exceeded the 0.7 threshold, implying satisfactory construct reliability (Fornell and Larcker, 1981; Dillon and Goldstein, 1984). The AVE values for each construct were all above the minimum critical value of 0.5, indicating good convergent validity for the measurement model (Chin *et al.*, 2003).

Discriminant validity was assessed using the Heterotrait–Monotrait Ratio (HTMT), which compares between-trait correlations and within-trait correlations. The HTMT approach, rather than classic approaches such as the Fornell-Larcker criterion, was adopted because HTMT is more effective and outperforms alternatives in terms of assessing discriminant validity (Henseler *et al.*, 2015). As Table 5 demonstrates, the HTMT value for each pair of constructs is below the 0.9 threshold for both the full sample and subgroups, suggesting good discriminant validity.

Next, a common method bias test was conducted to reinforce remedial approaches and prevent biases resulting from self-reported measurements. First, following Podsakoff and Organ (1986), a Harmon one-factor test was conducted. The results revealed that the variance explained by that single factor is 35.4%, below 50%, indicating that common method biases were not likely to contaminate the results. Second, following Liang *et al.* (2007), we conducted a statistical analysis by adding a common method factor that included all construct items as indicators. As Table 6 shows, most factor loadings were insignificant, with the substantive factor loadings remaining significant. Additionally, the average substantively explained indicator variance was 0.698, with the average method-based variance being 0.019. The ratio comparing substantive variance to method variance was around 37:1. Given the small magnitude and insignificant results for method variance, we concluded that common method bias was apparently not a serious concern.

4.2 Testing the structural model

To evaluate the research model and test the hypotheses, we adopted a PLS approach using SmartPLS 3.0. PLS and CB-SEM have proven equally effective for developing and analyzing structural relationships (Dash and Paul, 2021). However, according to Meri *et al.* (2019),

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Item	Category	Frequency	Ratio
Gender	Male	224	55.86%
	Female	177	44.14%
Age	Under 18	2	0.50%
	18–24	71	17.71%
	25–34	203	50.62%
	35–44	84	20.95%
	45–54	29	7.23%
	55 or above	12	2.74%
The highest education level	Below high school	0	0.00%
	High school	85	21.70%
	Bachelor's degree	215	53.62%
	Master's degree	99	24.69%
	PhD or above	0	0.00%
Employment	Employed full time	235	58.60%
	Employed part time	91	22.69%
	Unemployed	48	11.98%
	Retired	0	0.00%
	Student	27	6.73%

Table 3.
Demographic information ($N = 401$)

Construct	Items	Factor loadings	Cronbach's alpha	Composite reliability	rho_A	AVE
Perceived recommendation accuracy (PRA)	PRA1	0.904	0.810	0.887	0.848	0.725
	PRA2	0.904				
	PRA3	0.737				
Perceived recommendation serendipity (PRS)	PRS1	0.753	0.806	0.873	0.812	0.632
	PRS2	0.788				
	PRS3	0.827				
	PRS4	0.811				
Perceived effortlessness (PE)	PE1	0.821	0.749	0.881	0.939	0.787
	PE2	0.949				
Flow experience (FE)	FE1	0.767	0.833	0.887	0.859	0.663
	FE2	0.812				
	FE3	0.845				
	FE4	0.831				

Table 4.
Reliability and convergent validity analysis

compared with the CB-SEM approach, the PLS approach is suitable for incremental studies, i.e. constructing new measures and structural paths, particularly in IS research (Hair *et al.*, 2011). The current study not only tests the existing theories but also expands the theoretical framework by adding new paths (e.g. moderators). Therefore, PLS was deemed more appropriate.

Figure 2 and Table 7 present the results of the proposed model. As Figure 2 shows, flow experience is explained by 41.5% of the variation in the conceptual model. Because human behavior cannot be accurately predicted, most behavioral and social sciences studies that attempt to predict human behavior tend to produce R^2 values below 0.5. Falk and Miller (1992) proposed an R^2 value of 0.1 as a minimum acceptable level, and Chin (1998) suggested that PLS-derived R^2 values of 0.19, 0.33, and 0.67 can be considered weak, moderate, and

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substantial, respectively. According to these criteria, the endogenous latent variables produced acceptable R^2 values and can be considered to represent a moderate level of explanatory power.

Table 7 presents the hypothesis testing results in detail. As is demonstrated, user flow experience was significantly influenced by perceived recommendation accuracy ($\beta = 0.112$, $p < 0.05$) and perceived recommendation serendipity ($\beta = 0.419, p < 0.001$), supporting H1 and H2. Additionally, perceived effortlessness was significantly associated with flow experience ($\beta = 0.171, p < 0.001$), supporting H3. To test the moderating effects of user experience level and video length, we treated them the same as the other focal variables in the model and conducted the path analysis using the PLS-product indicator approach proposed by Chin *et al.* (2003). The results are shown in Table 7. The outcomes revealed that user experience level strengthens the effect of perceived recommendation serendipity ($\beta = 0.105, p < 0.05$) on flow experience. In addition, it weakens the effects of perceived recommendation accuracy ($\beta = -0.150, p < 0.01$) and perceived effortlessness ($\beta = -0.098, p < 0.05$) on flow experience. Thus, H4, H5, and H6 were supported by the results. Meanwhile, video length was found to weaken the effect of perceived effortlessness ($\beta = -0.117$, $p < 0.01$) on flow experience. Therefore, H7 was supported.

To get a broad picture of moderation patterns, we plotted the significant interactions with the Johnson and Neyman technique (1936), which reveals all ranges of the moderator values

	Full sample ($N = 401$)			Low-experience group ($N = 241$)			High-experience group ($N = 160$)		
	PRA	PRS	PE	PRA	PRS	PE	PRA	PRS	PE
PRA									
PRS	0.691			0.642			0.745		
PE	0.453	0.450		0.386	0.374		0.502	0.508	
FE	0.492	0.659	0.434	0.539	0.645	0.447	0.358	0.639	0.345

Note(s): PRA refers to perceived recommendation accuracy; PRS refers to perceived recommendation serendipity; PE refers to perceived effortlessness; FE refers to flow experience

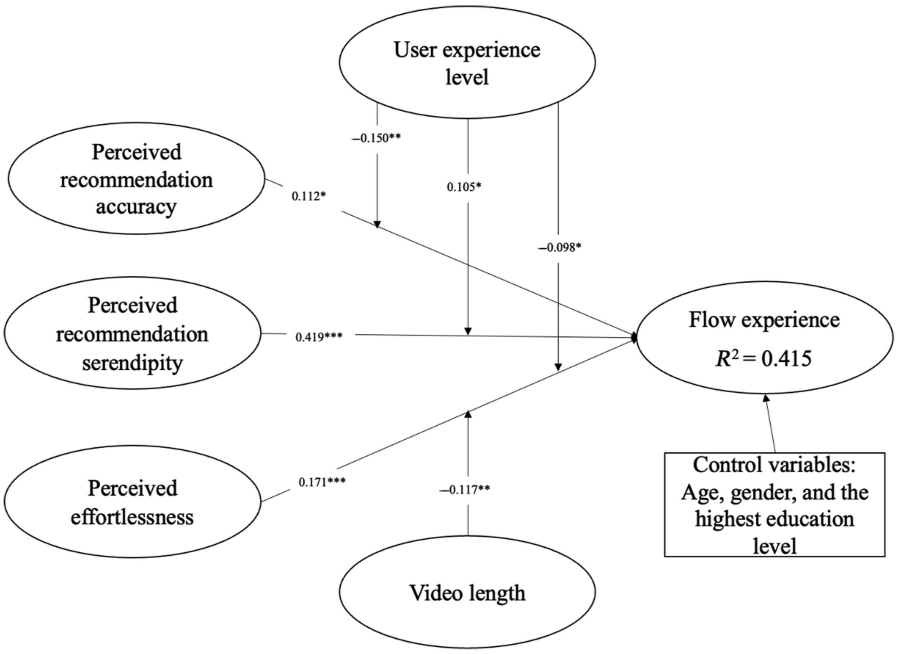
Table 5. HTMT analysis results

Construct	Indicator	Substantive factor loading (R1)	R1 square	Method factor loading (R2)	R2 square
Perceived recommendation accuracy (PRA)	PRA1	0.877***	0.769	0.016	0.000
	PRA2	0.852***	0.726	0.055	0.003
	PRA3	0.833***	0.694	−0.084	0.007
Perceived recommendation serendipity (PRS)	PRS1	0.820***	0.672	−0.073	0.005
	PRS2	0.869***	0.755	−0.083	0.007
	PRS3	0.805***	0.648	0.033	0.001
	PRS4	0.691***	0.477	0.117	0.014
Perceived effortlessness (PE)	PE1	0.971***	0.943	−0.094	0.009
	PE2	0.821***	0.674	0.139**	0.019
Flow experience (FE)	FE1	0.870***	0.757	−0.151	0.023
	FE2	0.993***	0.986	−0.233***	0.054
	FE3	0.546***	0.298	0.321***	0.103
	FE4	0.824***	0.679	0.002	0.000
Average		0.829	*0.698*	−0.003	*0.019*

Note(s): ** refers to $p < 0.01$; *** refers to $p < 0.001$; the numbers in italics refer to the average substantive variance and the average method variance

Table 6. Common method bias test results

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Note(s): *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Figure 2.
Structural model results

Hypotheses	β (standardized coefficient)	Standard errors	p-value	95% confidence interval Lower	Upper	Results
H1: PRA → FE	0.112*	0.054	0.037	0.003	0.218	Supported
H2: PRS → FE	0.419***	0.053	0.000	0.315	0.526	Supported
H3: PE → FE	0.171***	0.047	0.000	0.076	0.261	Supported
Moderation analysis						
UEL → FE	0.057	0.039	0.145	−0.019	0.135	
H4: PRA × UEL → FE	−0.150**	0.052	0.004	−0.252	−0.045	Supported
H5: PRS × UEL → FE	0.105*	0.051	0.040	0.002	0.200	Supported
H6: PE × UEL → FE	−0.098*	0.049	0.045	−0.191	−0.002	Supported
VL → FE	−0.056	0.043	0.189	−0.138	0.026	
H7: PE × VL → FE	−0.117**	0.040	0.004	−0.192	−0.034	Supported

Note(s): PRA refers to perceived recommendation accuracy; PRS refers to perceived recommendation serendipity; PE refers to perceived effortlessness; UEL refers to user experience level; VL refers to video length; FE refers to flow experience; ns refers to not significant, * refers to $p < 0.05$; ** refers to $p < 0.01$; *** refers to $p < 0.001$

Table 7.
Hypothesis testing results

where the relation of the predictor variable to the outcome variable is significant or insignificant. The results are shown in Figures 3–6. Figure 3 reveals that as user experience rises, the conditional effect on flow of perceived recommendation accuracy decreases. Specifically, as user experience increases above 2.93, the relationship between perceived

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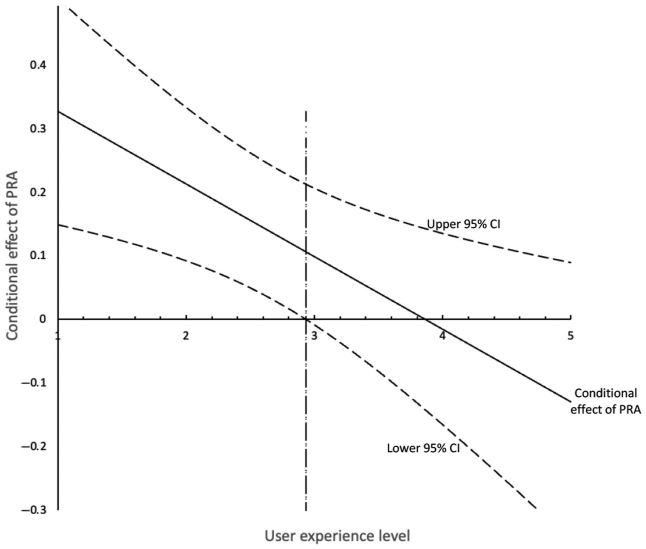


Figure 3.
Conditional effect of PRA on FE at varying levels of UE

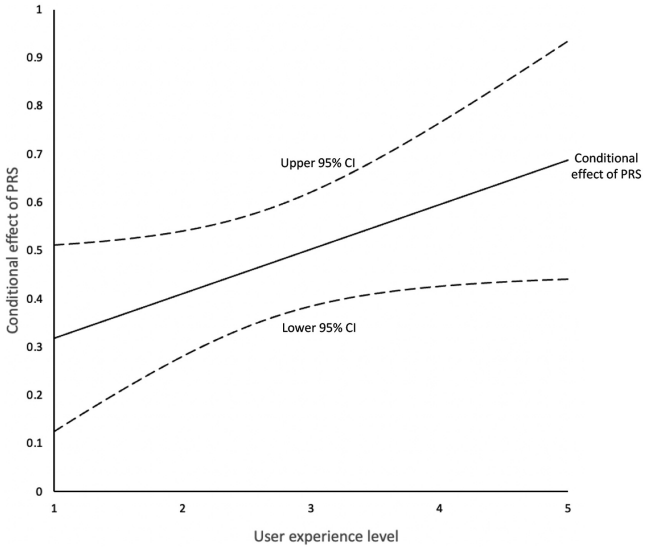


Figure 4.
Conditional effect of PRS on FE at varying levels of UE

recommendation accuracy and flow experience becomes insignificant. A similar interaction pattern was observed from perceived effortlessness on flow (see Figure 5). As expected, low levels of user experience led to perceived effortlessness more strongly influencing flow experience, with any positive significant effect disappearing for the users with user experience higher than 3.43. Besides, Figure 4 shows that the conditional effects of perceived recommendation serendipity were significantly positive for the total range of scores of user experience, with the conditional effects becoming stronger as user experience increases. Last,

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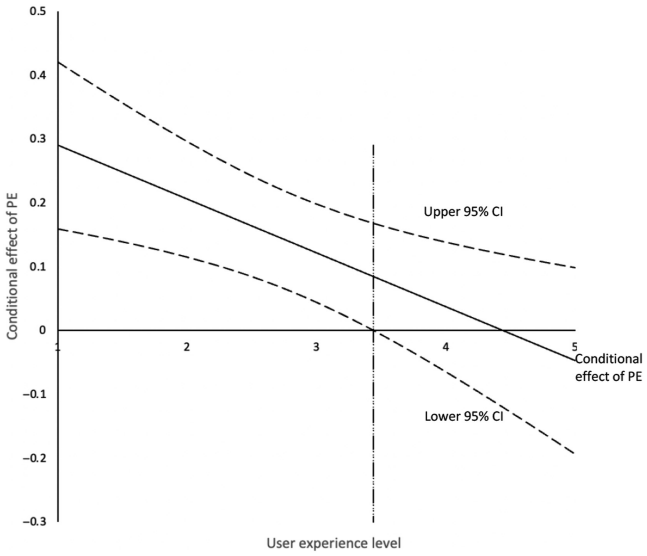


Figure 5.
Conditional effect of PE on FE at varying levels of UE

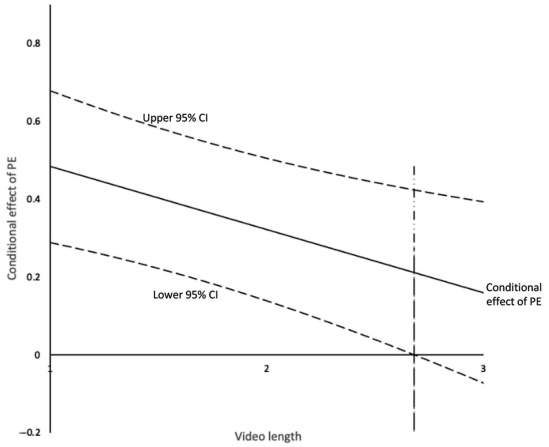


Figure 6.
Conditional effect of PE on FE at varying levels of VL

Note(s): PRA refers to perceived recommendation accuracy; PRS refers to perceived recommendation serendipity; PE refers to perceived effortlessness; UE refers to user experience; VL refers to video length; FE refers to flow experience; The black continuous lines show the effect of the focal predictor (PRA, PRS, and PE) on the outcome variable (FE) for different levels of the moderator (UE and VL), and the dotted lines above and below indicate the corresponding 95% confidence intervals (CI); The vertical dashed line indicates the Johnson and Neyman significance cut-off border

in terms of the moderating effect of video length, when the video length score is lower than 2.68, perceived effortlessness was found to positively influence users' flow experience and the conditional effect decreases as video length increases. When the video length is larger than 3.68, the relationship between perceived effortlessness and flow becomes insignificant (see Figure 6).

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4.3 A post hoc moderating effect test

To obtain a thorough understanding of the influence of user experience level, following the literature, we conducted a sub-group analysis in accordance with the procedure introduced by Keil *et al.* (2000). The literature classifies individuals who have used an SNS for more than 1 year as high-experience users and those who have used an SNS for less than 1 year as low-experience users (Lu and Lee, 2012; Shao *et al.*, 2020). Accordingly, we assigned those who had used TikTok for more than one year to the high-experience user group ($n = 241$) and those who had used TikTok for less than one year to the low-experience user group ($n = 160$). After dividing the dataset into these two user groups, a multi-group analysis was conducted to compare the path coefficient differences in the structural model for the two sub-samples, allowing observation of the moderating effect. This technique is recognized as valid for testing sub-group differences using PLS (Shao *et al.*, 2020; Silva *et al.*, 2019). Table 8 presents the results of the sub-group analysis.

As Table 8 indicates, perceived recommendation accuracy significantly influences user flow experience for the low-experience user group ($\beta = 0.188$, $p < 0.01$), with any similarly significant effect disappearing for the high-experience user group. This result implies that perceived recommendation accuracy is effective only for low-experience users. Additionally, following the calculation procedure provided by Keil *et al.* (2000), a multi-group analysis was conducted to compare the path coefficient differences in the structural models of the two sub-samples, with the results ($\beta_{low} = 0.188$, $\beta_{high} = -0.005$, $t_{spooled} = 24.45$) also providing empirical evidence for a substantial difference between the two subgroups, supporting H4. Meanwhile, the path coefficient from perceived recommendation serendipity to flow experience was significantly greater for the high-experience user group than the low-experience user group ($\beta_{low} = 0.393$, $\beta_{high} = 0.464$, $t_{spooled} = -9.85$), supporting H5. Finally, the relationship between perceived effortlessness and user flow experience was significant for the low-experience user group ($\beta = 0.209$, $p < 0.001$) but insignificant for the high-experience user group. Given that the statistical multi-group analysis also reported significant differences between the two groups ($\beta_{low} = 0.209$, $\beta_{high} = 0.116$, $t_{spooled} = 12.90$), H6 was supported. Figure 7 presents the subgroup analysis results for the high- and low-experience user groups.

5. Discussion and implications

5.1 Discussion

This study examined the effects of technology affordances, as moderated by user experience level and video length, on flow experience in the context of TikTok. Given the support for all the proposed hypotheses, this study provides several interesting insights.

First, although the strategies underscoring TikTok's capacity to engage users remain unclear, technology affordance theory provides some understanding of how TikTok's innovative and advanced technologies have contributed to user engagement, with three specific salient technology affordances (perceived recommendation accuracy, perceived

	Low-experience group ($N = 241$)	High-experience group ($N = 160$)	$t_{spooled}$
PRA → FE	0.188** ($p = 0.005$)	0.005 (ns) ($p = 0.956$)	24.45***
PRS → FE	0.393*** ($p = 0.000$)	0.464*** ($p = 0.000$)	−9.85***
PE → FE	0.209*** ($p = 0.000$)	0.116 (ns) ($p = 0.174$)	12.90***
R^2	0.392	0.276	

Note(s): PRA refers to perceived recommendation accuracy; PRS refers to perceived recommendation serendipity; PE refers to perceived effortlessness; FE refers to flow experience; ns refers to not significant, * refers to $p < 0.05$; ** refers to $p < 0.01$; *** refers to $p < 0.001$

Table 8. Path coefficient comparison between high- and low-experience users

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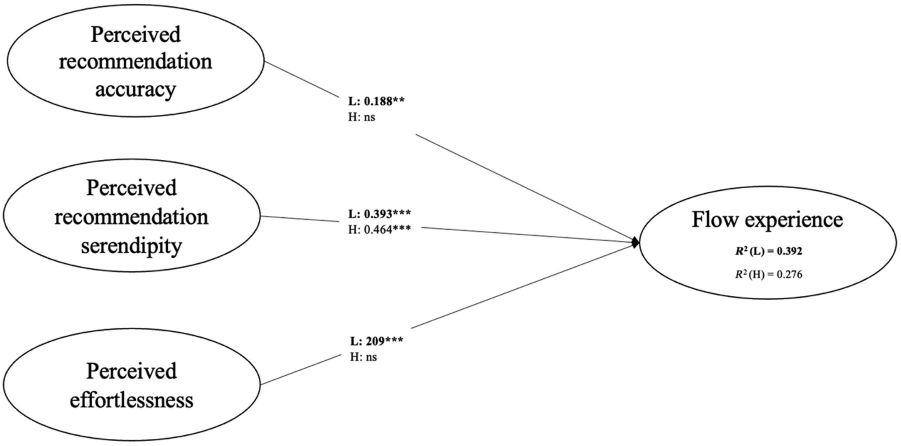


Figure 7.
Path coefficient comparison between low- and high-experience users

Note(s): *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; L refers to low-experience group results, and H refers to high-experience group results

recommendation serendipity, and perceived effortlessness) having been demonstrated to greatly impact user flow experience. Therefore, the current study reveals that TikTok's algorithm-run feed mechanism, which automatically delivers recommended short videos to users without requiring a cognitive self-directed selection process, represents one of the reasons for its substantial popularity. Meanwhile, given that perceived recommendation serendipity demonstrated the strongest effect ($\beta = 0.419$) on user flow experience, the unpredictability of recommended videos contributes most significantly to engaging users, with the innovative approach to content presentation allowing short-video platforms to stand out from traditional social media.

Next, although some studies have noted the relationship between technology affordances and flow experience in the IS field, many have neglected the impact of person-related factors on user experience level, leading to uncertainty regarding when serendipity should be introduced to a recommendation system or how user interest can be maintained in later usage stages. Our results indicate that when a user begins using a platform, all three technology affordances are significantly associated with optimal flow experience. However, the effects of perceived effortlessness and recommendation accuracy disappear for long-term users, who are more likely to be influenced by perceived serendipity. Thus, our results confirm that user experience level significantly shapes the effects of technology affordances. Specifically, while the influence of perceived recommendation serendipity is significantly stronger for high-experience users, perceived recommendation accuracy and perceived effortlessness more strongly influence low-experience users. This finding aligns with adaptation level theory. Although innovative and attractive technologies can hook users at the beginning, many users will adapt and demand new and unexpected experiences.

Finally, given the unique presentation of short-format videos, this study has also considered the role of video length in determining the effectiveness of technology affordances. Our results indicate that video length significantly weakens the impact of perceived effortlessness, suggesting that TikTok's success might be due to a combination of its unique technology and its short-format content.

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5.2 Theoretical implications

Our study makes four major theoretical contributions. First, adopting flow theory in the TikTok context, this study examines several critical antecedents to user flow experience. The literature signals the need to consolidate the research on flow in conjunction with context, especially as new technologies are introduced, and to develop deep knowledge about the state and stages of flow that builds on different perspectives (Valinatajbahnamiri and Siahtiri, 2021). As a relatively new and very popular application, TikTok successfully manipulates user flow experience. However, the underlying stimuli that contribute to this success remain under-investigated. Despite substantial progress in research on flow experience in the online environment (e.g. Trevino and Webster, 1992; Webster *et al.*, 1993; Hoffman and Novak, 1996), the applicability of these results in the TikTok context remains unclear. Hence, by specifically investigating user flow experience in the TikTok context, the current research extends this line of inquiry. Consistent with the literature, our findings reveal that artifact-related factors, person-related factors, and context-related factors constitute significant antecedents to user flow experience. Regarding artifact-related factors, the finding that perceived effortlessness significantly leads users to flow experience, consistent with existing findings (Harris *et al.*, 2017; Zhou, 2012), confirms its generalizability from an online environment to the TikTok context. However, the path from an algorithm's intelligent recommendation to flow experience remains largely unexplored, especially in terms of the relationship between perceived recommendation serendipity and flow experience. According to Lu and Cheng (2020), "Past studies on flow experience in the information field failed to discuss that serendipity also leads to the flow phenomenon." By revealing the specific effects of TikTok's recommendation mechanisms, the current study has identified recommendation-related affordances (i.e. perceived recommendation accuracy and perceived recommendation serendipity) as the main artifact-related factors leading to flow experience in the TikTok context, suggesting its smart recommendation algorithm represents the critical factor for the platform's substantial success. Given that perceived recommendation accuracy and perceived recommendation serendipity both demonstrated significant effects on flow experience, this research enriches flow theory and suggests the possibility of further research concerning the short-video user experience from the perspective of the TikTok recommendation algorithm.

Next, the current study has identified a significant impact of user experience level, a person-related factor, on user flow experience in the TikTok context. Although the extensive literature on flow often captures person-related factors as personal traits that interact with the artifact-related factors to determine user flow experience (e.g. Wu *et al.*, 2016; Kang *et al.*, 2018; Kim *et al.*, 2019; Jeon *et al.*, 2018), personality traits are relatively invariant to situational stimuli (Li and Browne, 2006), which undermines the capacity of platforms to detect the reasons for losing users. Furthermore, most previous studies neglected the possibility that the same activity may not induce a consistent flow experience as the person's experience with a technology increases. According to Valinatajbahnamiri and Siahtiri (2021), person-related factors that reflect dynamic individual states remain largely under-investigated. The literature calls for empirical evidence to identify the possible factors that contribute to maintaining user interest as a user's experience increases to prevent the loss of users. Thus, by capturing the critical person-related factor of user experience level, the study extends flow theory. Furthermore, our results encourage future studies to pay attention to dynamic individual state factors, especially the factors that contribute to retaining users as their experience level increases, with this study's results showing that perceived recommendation serendipity might represent a key factor.

Third, the current study contributes to the recommendation system research by evaluating the recommendation algorithm from the user's perspective. Most studies of recommendation systems are design science studies that introduce complex algorithms to promote the accuracy of recommendations (Wang *et al.*, 2020; Roudposhti *et al.*, 2018;

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Zhu *et al.*, 2018). Following the appearance of problems such as the "filter bubble" and "information cocoons," recommendation serendipity has become a sought-after property, with many researchers from the design science field introducing advanced algorithms to inject serendipity into recommendation systems (Wang *et al.*, 2020; Cai *et al.*, 2021). Thus, design science persists as the main approach to investigating recommendation systems, with relevant behavioral studies scant. This has precluded a comprehensive understanding of user preferences in terms of recommendations. The current study not only identifies user preferences in terms of recommendation accuracy and serendipity but also reveals variations between high- and low-experience users. By adopting user experience level as a moderator, our study demonstrates that the effect of recommendation accuracy significantly weakens—and the effect of recommendation serendipity substantially increases—as user experience level increases. Thus, by collecting first-hand data from users, our study enriches the literature on recommendation systems, encouraging further studies concerning user preferences when designing such systems.

Finally, our study introduces a context-dependent moderator called video length to moderate the effects of technology affordances, a first in the short-video platform context. The results indicate that video length significantly shapes the effect of perceived effortlessness on optimal user flow experience. That is, shorter content means effortlessness more substantially determines optimal user flow experience. Thus, this context-dependent moderator can both explain the influence of technology on the TikTok user experience and inspire future researchers to consider the unique content presentation format when investigating user behavior in the short-video platform context.

5.3 Practical implications

The current study provides practitioners with various insights regarding building sustainable short-video platforms. First, our results indicate that it is critical to be aware of how different technologies shape user experience, with both the recommendation systems and an effortless operating process improving the user flow experience. This finding provides several insights for short-video platform managers. In terms of the role of perceived effortlessness, the simpler the operation process, the better the user experience. A critical reason for TikTok's popularity is its extremely simple operation process (a simple swipe enables the operation of most functions, and the content is fed to users automatically). Thus, this study's results indicate that short-video application designers should be aware of the impact of effortlessness on user experience. For example, China's second-largest short-video platform, Kwai, initially embedded a lengthy suggestion list (rather than automatically displaying content), which produced extra cognitive load. Many users reported disliking this design, preferring the automatic content-feeding practice of the Chinese version of TikTok. This practice shows that users prefer a "don't make me think" (i.e. perceived effortlessness) operating style. Although short-video platforms introduce new functions to maintain user interest as they develop, some functions may increase the effort required and negatively affect user experience. Thus, when introducing new functions, short-video platform practitioners should be aware of the problems associated with system-feature overload, which can fatigue users.

Second, although extremely simple and automatic content-feeding mechanisms benefit short-video platforms, in this case, the recommendation system more significantly determines user experience. Thus, our study encourages practitioners to focus on two dimensions of recommendation technologies: recommendation accuracy and recommendation serendipity. Although contemporary recommendation algorithms can provide content that very precisely matches user preferences, this does not mean that more accurate recommendations necessarily benefit users. Many short-video platforms, as well as

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other recommendation-oriented platforms, place too much emphasis on improving their algorithm's accuracy, which can easily produce the "information cocoon" trap. By examining recommendation preferences from the user perspective rather than adopting the design science approach, the current study provides insights for recommendation systems to invest more effort in improving serendipity-related factors rather than only improving the precision of the algorithm. Notably, our results reveal that the effectiveness of recommendation accuracy and serendipity depend on a user's experience level. Although previous studies have indicated the contribution of serendipity to solving the overfitting problem of recommendation systems, there remains no comprehensive understanding of how and when serendipity impacts users. Considering our results demonstrate that the effects of perceived recommendation accuracy decrease and the effects of recommendation serendipity increase as user experience level increases, this study encourages designers to incorporate awareness of user experience level into recommendation algorithms.

By introducing video length as a moderator, it is possible to understand different user reactions to technology affordances based on different video lengths, which might provide some guidance to short-video platform practitioners. For example, certain technology affordances, especially perceived effortlessness, work only with shorter-format content. In the context of TikTok's gamble on users preferring longer-format videos, our results constitute a forewarning about the risk associated with the shift. Allowing longer videos risks depriving these platforms of the uniqueness of short videos—compared to longer-video platforms (e.g. YouTube). Hence, the influence of video length on user experience is worth exploring further.

Beyond short-video platforms, our study may provide insights for other platform types. First, given that the challenge of maintaining user interest is faced by many mature online communities (Fu, 2019), by employing experience level as a moderator, our results indicate that online communities should be aware that the factors that attract users initially might differ from the factors associated with retaining users. The general principle (i.e. user preferences vary according to experience level) is of interest to most online community managers. Practices such as introducing surprises and unexpected experiences may contribute to maintaining long-term user engagement, with the specific effects requiring further study for verification in different contexts. Second, although TikTok's design principles have enabled substantial success for the platform, we advise longer-format video platforms (e.g. YouTube) to avoid copying its operating mechanisms. Some aspects may function better for short-format video platforms than for longer-format video platforms, especially aspects associated with simplification. Consider, for example, if YouTube removed its suggestion feature and simply began to reproduce videos the moment a user started browsing, doing so might not positively impact user experience because perceived effortlessness plays a much more significant role in the context of short-format video platforms. Thus, given that TikTok's success might be due to a combination of its unique technology and its short-format content, it is necessary to further investigate whether TikTok's design principles can be applied successfully by longer-format video platforms.

5.4 Limitations and recommendations for future research

The current study features several limitations. First, the limited variation in video length that characterizes TikTok meant that the study scope precluded consideration of how video length moderates the relationship between algorithm-related factors and flow experience. Previous studies have suggested that users are sensitive to content type only when video length varies to a substantial degree (Guo *et al.*, 2014; Verma *et al.*, 2021). Therefore, there are limited insights regarding whether the recommendation algorithm functions similarly in the long-video context. Future studies should capture the perspectives of users of short- and long-

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form video platforms to fully understand the role that video length plays. For example, researchers could compare the preferences of users of long-form video platforms (e.g. YouTube) and users of short-form video platforms (e.g. TikTok) to explore how video length impacts cognitive responses. By considering videos longer than 3 min, it might be possible to identify the video-length range that is optimal for user flow experience. Such insights would also be valuable for TikTok, with the company currently testing the outcomes of allowing longer videos (up to 15 min).

Second, our study's moderate R^2 value, according to the criteria introduced by Chin (1998), prompts the suggestion that future studies adopt other theoretical lenses to identify other important factors that could contribute to increased flow in the TikTok context. In terms of artifact-related factors, perceived privacy, aesthetic design, and interface design (e.g. horizontal vs. vertical) could all potentially affect user flow experience on TikTok. For person-related factors, future studies could examine the static factors that determine user flow experience in this context, such as individual personality, self-control, perceived innovativeness, and novelty-seeking. Finally, we encourage future studies to capture other dimensions when investigating the user experience on TikTok. These other dimensions might include environment-related factors, task-related factors, and possibly cultural differences.

6. Conclusions

This paper's findings enable the conclusion that TikTok's technology affordances cannot be overlooked when investigating the platform's global rise, with our results demonstrating that three technology affordances (perceived recommendation accuracy, perceived recommendation serendipity, and perceived effortlessness) significantly explain optimal user flow experience in the context of short-video platforms. Interestingly, all three technologies significantly affect flow experience for low-experience users. However, perceived recommendation serendipity is critical to promoting flow experience among high-experience users. Furthermore, the effect of perceived effortlessness on flow experience is significantly stronger for shorter videos than for longer videos. Ultimately, this study demonstrates that these three perceived technology affordances are moderated by user experience level and video length to impact optimal user flow experience in the short-video platform context.

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Appendix

Constructs	Items	Sources
Perceived recommendation accuracy	PRA1: On TikTok, the recommended videos match my interests PRA2: On TikTok, the videos recommended to me match my preferences PRA3: On TikTok, each of the recommended videos is relevant	Roudposhti *et al.* (2018), Knijnenburg *et al.* (2012)
Perceived recommendation serendipity	PRS1: I can often see unexpected content in the TikTok videos that I am interested in or are helpful to me, and I have a feeling of surprise PRS2: I am often surprised by the TikTok recommended videos that I never found, and they are interesting and helpful to me PRS3: I often find interesting and surprising content in the TikTok videos PRS4: There is a lot of valuable content, more than what I want to know when I use TikTok	Lu and Cheng (2020)
Perceived effortlessness	PE1: Using TikTok is easy PE2: Using TikTok is effortless	Yu and Kong (2016)
Flow experience	FE1: I often lose track of time and immerse in browsing and watching TikTok FE2: I often concentrate and ignore other things when I browse and watch TikTok videos FE3: I often feel fascinated when I browse and watch TikTok videos FE4: I am very dedicated and often feel a distorted sense of time when I browse and watch TikTok videos	Lu and Cheng (2020)
User experience level	How long have you been using TikTok? 1. less than 4 months; 2. 4–8 months; 3. 8–12 months; 4. 1–2 years; 5. more than 2 years	Shao *et al.* (2020)
Video length	The videos that are recommended to me on TikTok are usually: 1. less than 15 s; 2. 15–60 s; 3. 1–3 min; 4. more than 3 min	Lin *et al.* (2020)

Table A1.
Measurement items

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